not the case): take, for example, the randomized version of Bogosort.

F.4 Classification Metrics

When we analyze data structures like Bloom filters or Treaps, examining the time and memory requirements of their methods is paramount, but not enough: besides how fast they run, and how much memory they use, for Monte Carlo algorithms we also need to ask one more question: “how well does it work”?

To answer this question, we need to introduce *metrics*, functions that measure the distance between an approximated solution and the optimal one.

For classification algorithms, the quality of an algorithm is a measure of how well each input is assigned to its right class. For binary (i.e. true/false) classification, therefore, it's about how accurately the algorithm outputs true for the input points.

F.4.1 Accuracy

One way of measuring the quality of a classification algorithm is to assess the rate of correct predictions. Suppose that over N_P points actually belonging to the `true` class

- P_P are predicted as `true`